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Mining Toxicity Information from Large Amounts of Toxicity Data

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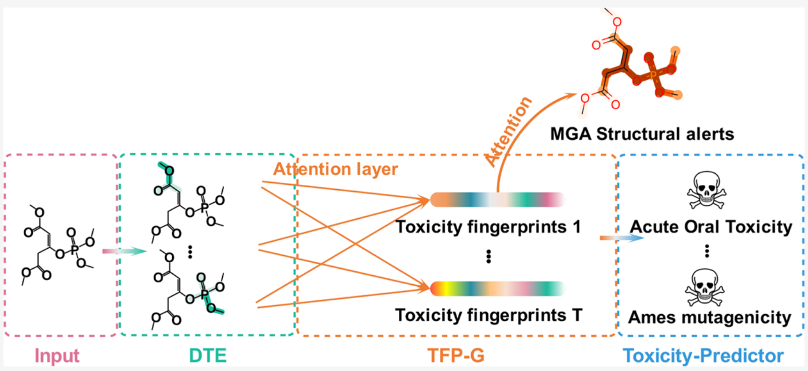

ABSTRACT: Safety is a main reason for drug failures, and therefore, the detection of compound toxicity and potential adverse effects in the early stage of drug development is highly desirable. However, accurate prediction of many toxicity endpoints is extremely challenging due to low accessibility of sufficient and reliable toxicity data, as well as complicated and diversified mechanisms related to toxicity. In this study, we proposed the novel multitask graph attention (MGA) framework to learn the regression and classification tasks simultaneously. MGA has shown excellent predictive power on 33 toxicity data sets and has the capability to extract general toxicity features and generate customized toxicity fingerprints. In addition, MGA provides a new way to detect structural alerts and discover the relationship between different toxicity tasks, which will be quite helpful to mine toxicity information from large amounts of toxicity data.

1. INTRODUCTION

During drug discovery and development, unexpected toxicities are the main factor that impedes potential drug candidates into the market.[1] It was reported that drug safety issues are responsible for the attrition of around 30% of drug candidates and even lead to costly withdrawal of many marketed drugs.[2] Therefore, the detection of potential compound toxicities and side effects in the early stage of drug development is highly desirable. However, conventional in vivo and in vitro experimental approaches for toxicity evaluation are time-consuming and costly.[3,4] With the rapid growth in the amount of toxicological data, the development of time- and cost-efficient in silico methods for accurate toxicity prediction has gained increasing attention.[5−7]

In the past decade, a high number of toxicity prediction models with acceptable accuracy have been developed through various machine learning (ML) algorithms.[8−11] However, many challenges still need to be overcome before reliable prediction of toxicities can be achieved.[5] The first challenge is the low accessibility of sufficient and reliable toxicity data. Despite the recent rise in both volume and variety of toxicity data, it is still difficult to retrieve and combine the experimental data from different sources due to poor annotation and/or lack of accessibility. The existing data in the public domain for some important toxicity endpoints, such as urinary tract toxicity, only contain a few hundreds of molecules, which may not be enough for ML to construct robust prediction models.[12] Considering the existence of potential relationships between different toxicity endpoints, multitask learning (MTL) and transfer learning (TL) may be good solutions by making full use of the information learned from the related toxicity data.[13−15] The inspiration of MTL is that we can use the experience of related tasks to help learn new tasks, and the purpose of MTL is to improve the prediction accuracy by learning multiple related tasks jointly.[13,16] As a general learning paradigm, MTL has been widely used in computer vision,[17] bioinformatics,[18,19] speech,[20] and natural language processing,[21] and naturally, it has also been implemented into toxicity prediction.[22,23] The basic ideas of TL and MTL are similar. The significant difference is that TL aims at improving the performance of target tasks, while MTL aims at improving the

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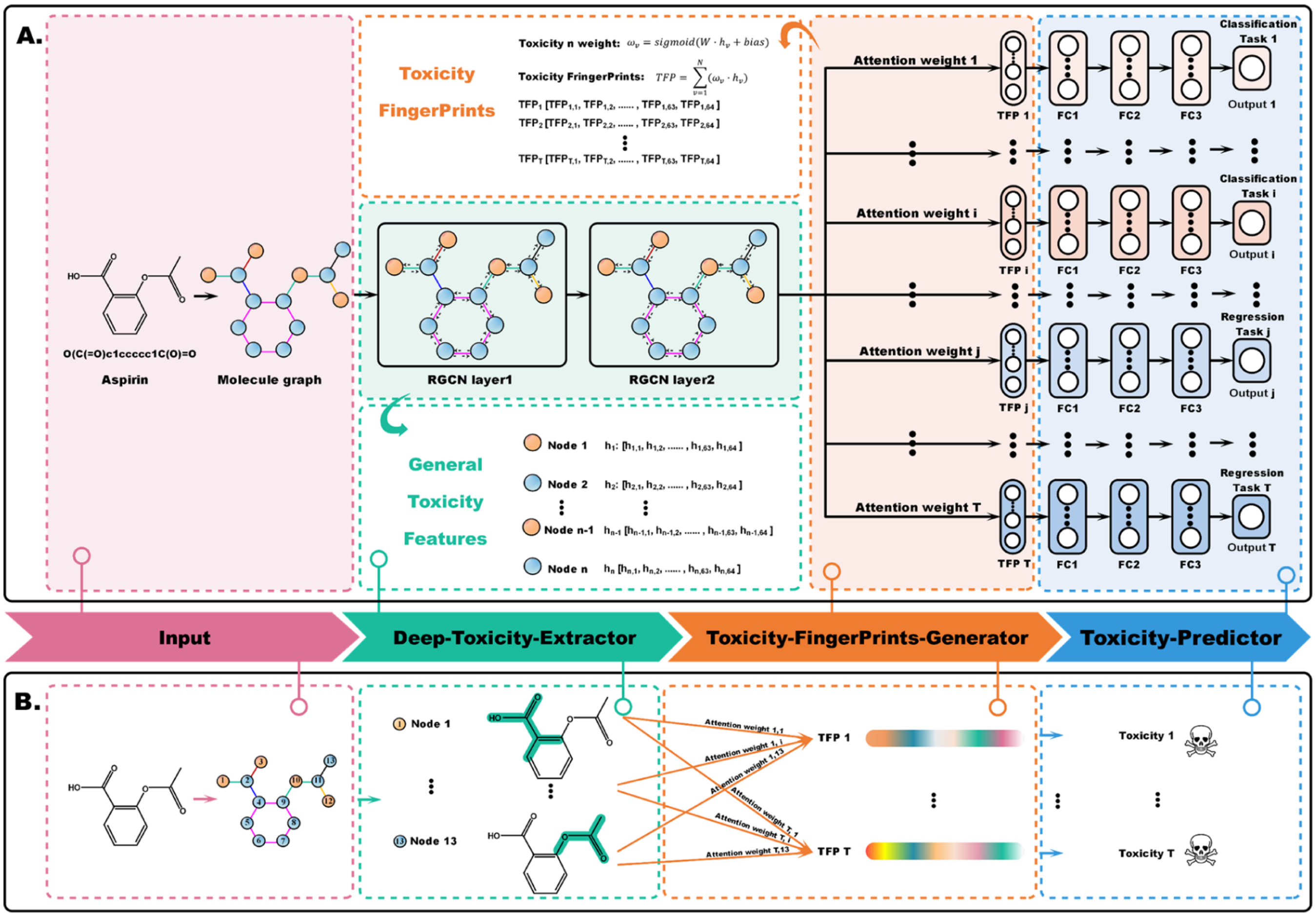

Figure 1. (A) Overview of the MGA framework. (B) Example of MGA for toxicity prediction.

performance of all tasks.[13] Therefore, MTL and TL will help us to solve the problems of insufficient toxicity data and low prediction accuracy of toxicity.

How to interpret the relationships between substructures and chemical toxicities is another challenge for toxicity prediction. Structural alerts (SAs), which are defined as the key substructures responsible for a certain toxicity, are usually used to reveal these relationships. However, SAs that represent numerous but redundant substructures in terms of their frequency of occurrence may not fully characterize the chemical or biological mechanisms related to a certain toxicity.[5] In recent years, graph neural networks (GNNs) have attracted increasing attention in molecular property prediction due to its strong predictive power and interpretability, which may provide a new way to reveal the relationships between substructures and chemical toxicities.[24−29] In addition, the combination of attention mechanism and GNN may provide new clues for SAs.

In this study, we proposed multitask graph attention (MGA), a novel framework to make full use of available toxicity data, which can simultaneously learn the regression and classification tasks for toxicity prediction. The results also demonstrate that MGA can be of great help in solving the abovementioned toxicity prediction problems that cannot be well-addressed by existing methods. Since making full use of existing data sets can be a way to "*improve the quality of a data set*", MGA may provide a solution to insufficient toxicity data. In order to verify whether MGA can make full use of data through MTL and improve the predictive ability of the model, we evaluated the predictions of the single tasks and multitasks of MGA to the 31 toxicity data sets. Compared with the single tasks, the multitasks can yield much better predictions for 17 tasks and worse prediction for only one task. In addition, we found that MGA can indeed extract the general toxicity features for specific substructures rather than a black box that cannot be easily understood, and these toxicity features can be transferred to new toxicity tasks through TL. Moreover, MGA can generate the customized toxicity fingerprints from the general toxicity features toward specific toxicity tasks. These toxicity fingerprints can also be used by other ML methods, not just neural networks, to develop robust toxicity predictors. Moreover, MGA provides a method to reveal the relationships between substructures and chemical toxicities by assigning different weights to different substructures, and these weights can be used as the signatures to highlight the SAs for specific tasks. Therefore, MGA provides a powerful framework for medicinal chemists to mine complicated toxicity information from large amounts of toxicity data.

2. RESULTS AND DISCUSSION

2.1. Performance of MGA and RGCN on the External Test Sets. Since the relation graph convolution network (RGCN) layers are used in MGA, we compared the performance of MGA (multitask) and RGCN (single task). A total of 31 data sets covering 8 toxicity categories were used to develop the multitask MGA models (Figure 1 and Table 1)

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Table 1. Atom and Bond Features Used in MGA

atom feature	size	description
atom symbol	16	[B, C, N, O, F, Si, P, S, Cl, As, Se, Br, Te, I, At, metal] (one-hot)
degree	7	number of covalent bonds [0,1,2,3,4,5,others] (one-hot)
formal charge	1	electrical charge (integer)
radical electrons	1	radical electrons 1 number of radical electrons (integer)
hybridization	6	[sp, sp2, sp3, sp3d, sp3d2, other] (one-hot)
Aromaticity	1	whether the atom is part of an aromatic system [0/1] (one-hot)
hydrogens	5	number of connected hydrogens [0,1,2,3,4] (one-hot)
chirality	1	whether the atom is chiral center [0/1] (one-hot)
chirality type	2	[R, S] (one-hot)
bond feature	**size**	**description**
bond type	4	[single, double, triple, aromatic] (one-hot)
conjugation	1	whether the bond is conjugated [0/1] (one-hot)
ring	1	whether the bond is in ring [0/1] (one-hot)
stereo	4	[StereoNone, StereoAny, StereoZ, StereoE] (one-hot)
atom pair	12	[['CC'], ['CN', 'NC'], ['ON', 'NO'], ['CO', 'OC'], ['CS', 'SC'],['SO', 'OS'], ['NN'], ['SN', 'NS'], ['CCl', 'ClC'], ['CF', 'FC'],['CBr', 'BrC'], ['others']] (integer)

and single-task RGCN models, and the hepatotoxicity and urinary tract toxicity data sets were used to assess whether the general toxicity features learned by MGA are transferable (Table 2). We performed 10 independent runs with different random seeds to train the models, and the performances of MGA and RGCN are shown in Table 3.

2.1.1. Carcinogenicity and Mutagenicity. Chemical carcinogenicity is of great concern in drug development due to its hazards to human health. Many approved drugs have been withdrawn from the market because they have been identified as carcinogens in humans or animals.[30] Therefore, many computational approaches have been developed in recent years to predict the carcinogenicity of compounds in the early stage of drug development.[31−34] The carcinogenic potency database (carcinogenicity) and Ames mutagenicity data sets are two commonly used data sets for the development of the prediction models and SAs for chemical carcinogenic toxicity.[34,35] As shown in Table 3, MGA significantly outperforms RGCN on the carcinogenicity data set and achieves an average area under the receiver operating characteristic curve (ROC-AUC) of 0.793. As to the Ames mutagenicity data set, the performances of MGA and RGCN are similar (ROC-AUC = 0.853 for MGA and ROC-AUC = 0.828 for RGCN), and there is no significant difference between them (P = 0.080).

2.1.2. Acute Oral Toxicity. Acute oral toxicity is one of the most concerned issues for drug development, and the drug candidates with high acute toxicity will not be considered further.[23,36] As shown in Table 3, RGCN shows slightly better performance than MGA in acute oral toxicity prediction, but their difference is not statistically significant (P = 0.239).

2.1.3. Cardiotoxicity. Undesirable hERG-related cardiotoxicity is one of the main reasons for drug candidate failures and drug withdrawal from the market.[37,38] It should be noted that there is lack of a specific threshold to discriminate hERG blockers from nonblockers,[39] and therefore, the data sets with four IC_{50} thresholds (1, 5, 10, and 30) were used for model building. As shown in Table 3, the performances of MGA and

Table 2. Detailed Information about the 33 Toxicity Data Sets

category	toxicity endpoint	model	no.
carcinogenicity and mutagenicity	carcinogenicity	classification	1042
	Ames mutagenicity	classification	7672
acute oral toxicity	LD_{50}	regression	5220
cardiotoxicity	cardiotoxicity-1	classification	1565
	cardiotoxicity-5	classification	1565
	cardiotoxicity-10	classification	1565
	cardiotoxicity-30	classification	1565
respiratory toxicity	respiratory toxicity	classification	1399
irritation and corrosion	EI	classification	5220
	EC	classification	2298
CYP450	CYP1A2	classification	9748
	CYP2C19	classification	10598
	CYP2C9	classification	9906
	CYP2D6	classification	10682
	CYP3A4	classification	11369
endocrine disruption	NR-AR	classification	7245
	NR-AR-LBD	classification	6740
	NR-AhR	classification	6529
	NR-aromatase	classification	5804
	NR-ER	classification	6176
	NR-ER-LBD	classification	6936
	NR-PPAR-gamma	classification	6432
	SR-ARE	classification	5816
	SR-ATTAD5	classification	7054
	SR-HSE	classification	6447
	SR-MMP	classification	5795
	SR-p53	classification	6756
Eco-toxicity	LC50DM	regression	347
	BCF	regression	676
	LC50	regression	816
	IGC50	regression	1787
hepatotoxicity	hepatotoxicity	classification	1313
urinary tract toxicity	urinary tract toxicity	regression	233

RGCN are quite close for the thresholds of 1, 5, and 30 μM. However, MGA significantly outperforms RGCN for the threshold of 10 μM and achieved the average ROC-AUC of 0.671.

2.1.4. Respiratory Toxicity. Respiratory toxicity is a widespread occupational and environmental problem in many industrializing countries. In addition, drug-induced respiratory toxicity is often underdiagnosed because it may not have obvious early signs or symptoms in general drug therapy.[40−42] Therefore, the prediction of respiratory toxicity is of great importance. As shown in Table 3, MGA illustrates stronger predictive power than RGCN in respiratory toxicity prediction.

2.1.5. Irritation and Corrosion. Although eye irritation (EI) and eye corrosion (EC) might not be considered in the drug discovery stage, they still play an important role in pharmaceutical and cosmetics industries.[43,44] Substances that may cause EI and EC should be detected, such as the ocular pharmaceuticals.[45] As shown in Table 3, both MGA and RGCN show strong predictive power in the predictions of EI and EC, and RGCN performs slightly better than MGA.

2.1.6. CYP450. The inhibition of CYP450 enzymes may lead to decreased elimination and/or changed metabolic pathways of their substrates, which is the major cause of adverse drug−

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Table 3. Performances of MGA and RGCN on the External Test Sets

category	toxicity endpoint	metric	RGCN	MGA	IPV[a]	P[b]
carcinogenicity and mutagenicity	carcinogenicity	ROC-AUC	0.750 ± 0.032	0.793 ± 0.027	**0.043**	**0.006**
	Ames mutagenicity	ROC-AUC	0.828 ± 0.024	0.853 ± 0.009	0.025	0.080
acute oral toxicity	LD_{50}	R^2	0.624 ± 0.007	0.618 ± 0.013	−0.006	0.239
cardiotoxicity	cardiotoxicity-1	ROC-AUC	0.701 ± 0.017	0.694 ± 0.014	−0.007	0.332
	cardiotoxicity-5	ROC-AUC	0.630 ± 0.011	0.632 ± 0.019	0.002	0.873
	cardiotoxicity-10	ROC-AUC	0.619 ± 0.011	0.671 ± 0.024	**0.052**	**0.000**
	cardiotoxicity-30	ROC-AUC	0.652 ± 0.027	0.648 ± 0.012	−0.004	0.732
respiratory toxicity	respiratory toxicity	ROC-AUC	0.866 ± 0.016	0.881 ± 0.011	**0.015**	**0.036**
irritation and corrosion	EI	ROC-AUC	0.981 ± 0.003	0.979 ± 0.002	−0.002	0.124
	EC	ROC-AUC	0.986 ± 0.002	0.981 ± 0.003	−0.005	0.008
CYP450	CYP1A2	ROC-AUC	0.938 ± 0.007	0.949 ± 0.007	**0.011**	**0.003**
	CYP2C19	ROC-AUC	0.776 ± 0.017	0.795 ± 0.009	**0.019**	**0.011**
	CYP2C9	ROC-AUC	0.752 ± 0.014	0.751 ± 0.018	−0.001	0.897
	CYP2D6	ROC-AUC	0.834 ± 0.011	0.878 ± 0.014	**0.044**	**0.000**
	CYP3A4	ROC-AUC	0.918 ± 0.006	0.914 ± 0.006	−0.004	0.165
endocrine disruption	NR-AR	ROC-AUC	0.775 ± 0.014	0.736 ± 0.016	**−0.039**	**0.000**
	NR-AR-LBD	ROC-AUC	0.831 ± 0.020	0.828 ± 0.024	−0.003	0.732
	NR-AhR	ROC-AUC	0.878 ± 0.008	0.883 ± 0.006	0.005	0.211
	NR-aromatase	ROC-AUC	0.738 ± 0.018	0.826 ± 0.018	**0.088**	**0.000**
	NR-ER	ROC-AUC	0.700 ± 0.011	0.729 ± 0.016	**0.029**	**0.000**
	NR-ER-LBD	ROC-AUC	0.814 ± 0.016	0.860 ± 0.020	**0.046**	**0.000**
	NR-PPAR-gamma	ROC-AUC	0.707 ± 0.029	0.741 ± 0.027	**0.037**	**0.019**
	SR-ARE	ROC-AUC	0.818 ± 0.013	0.835 ± 0.011	**0.017**	**0.008**
	SR-ATTAD5	ROC-AUC	0.806 ± 0.023	0.889 ± 0.018	**0.083**	**0.000**
	SR-HSE	ROC-AUC	0.824 ± 0.014	0.832 ± 0.014	0.008	0.271
	SR-MMP	ROC-AUC	0.907 ± 0.007	0.923 ± 0.004	**0.016**	**0.000**
	SR-p53	ROC-AUC	0.837 ± 0.010	0.870 ± 0.009	**0.033**	**0.000**
eco-toxicity	LC50DM	R^2	0.404 ± 0.032	0.648 ± 0.030	**0.244**	**0.000**
	BCF	R^2	0.721 ± 0.020	0.725 ± 0.027	0.004	0.710
	LC50	R^2	0.659 ± 0.021	0.712 ± 0.023	**0.053**	**0.000**
	IGC50	R^2	0.809 ± 0.013	0.827 ± 0.013	**0.018**	**0.010**

[a]The improvement of MGA over RGCN. [b]The *P* values for the predictions of RGCN and MGA provided by the independent samples *t*-test. The IPV value higher than 0.01 and the *P* value less than 0.05 represent significant improvement (highlighted in bold). The IPV value less than −0.01 and the *P* value less than 0.05 represent significant reduction (highlighted in red). The other cases represent no significant change.

drug interactions.[46−48] Several drugs were withdrawn from the market due to adverse drug−drug interactions caused by their inhibition of CYP450, such as mibefradil and cerivastatin.[49,50] Therefore, accurate prediction of CYP450 inhibition is highly demanded. As shown in Table 3, MGA outperforms RGCN on CYP1A2, CYP2C19, and CYP3D6, and MGA and RGCN perform similarly on CYP2C9 and CYP3A4.

2.1.7. Endocrine Disruption. Endocrine disrupting chemicals (EDCs) interacting with nuclear receptors (NRs), such as estrogen receptors (ERs) and androgen receptors (ARs), may interfere with the normal functions of these endogenous steroid hormones and lead to adverse health consequences.[51−53] The Tox21 project provides a large amount of data for the prediction of potential chemical−NR interactions.[54] In this study, we developed the models based on all the 12 endpoints in Tox21. As shown in Table 3, the performances of MGA and RGCN are similar on NR-AR-LBD, NR-AhR, and SR-HSE. MGA only performs worse than RGCN on NR-AR while significantly outperforms RGCN on the remaining 8 endpoints.

2.1.8. Eco-toxicity. Drugs and their metabolites exposed to the environment may affect the ecosystem due to their bioactivity to creature.[55] Four eco-toxicity data sets from the Toxicity Estimation Software Tool software (TEST) were collected to build the prediction models.[56] The data sets include 96 h fathead minnow LC_{50} (LC50), 48 h *Daphnia magna* LC_{50} (LC50DM), 40 h *Tetrahymena pyriformis* IGC_{50} (IGC50), and bioconcentration factor (BCF). As shown in Table 3, MGA significantly outperforms RGCN on LC50DM, LC50, and IGC50 and performs similarly to RGCN on BCF.

As mentioned above, MGA shows much higher predictive capability than RGCN in toxicity prediction. The predictions of 17 toxicity endpoints have clearly benefited from MGA, while only the prediction of NR-AR becomes obviously worse (Figure 2). Besides, the overall performances of MGA and RGCN on the remaining 13 tasks are close. The results demonstrate that compared with learning toxicity tasks individually, learning multiple toxicity tasks jointly can improve the performance of the toxicity predictor. Next, we will verify whether MGA can extract toxicity features and predict chemical toxicity according to our original design.

2.2. Does MGA Extract Toxicity Features and Predict Chemical Toxicity According to Our Original Design? In our original design, MGA can be divided into four components according to their functions: input, deep toxicity extractor (DTE), toxicity fingerprint generator (TFP-G), and toxicity predictor. DTE is supposed to extract the general toxicity features of the substructures from the initial atom and bond features. TFP-G is supposed to assign different attention weights to substructures and then generate the customized

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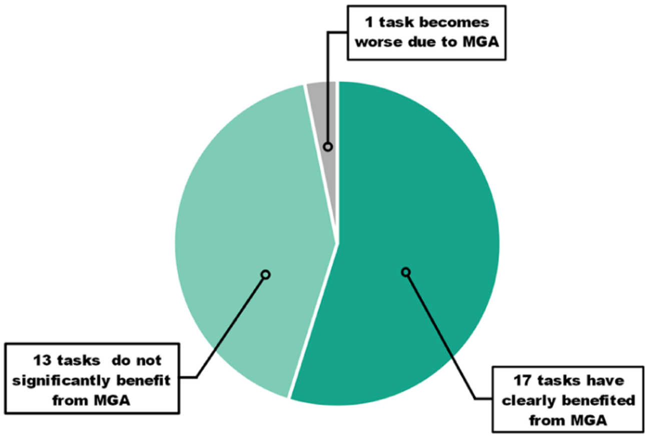

Figure 2. Comparison of the performances between MGA and RGCN.

toxicity fingerprints from the general toxicity features toward a specific toxicity task. In the following sections, we will explore whether DTE and TFP-G can achieve their functions as we have designed.

2.2.1. What Does DTE Learn? DTE is composed of two graph convolutional network (GCN) layers. As proved by David Duvenaud et al., the GCN can be considered as generalized molecular feature extraction methods.[57] The node features extracted by GCN are differentiable fingerprints to capture the chemical information of circular substructures, which can be regarded as a general form of extended connectivity fingerprints (ECFPs).[57] To ensure that DTE can extract the general toxicity features of circular substructures, we used 31 toxicity data sets to simultaneously train the model and make DTE become the only shared part of the model. Therefore, the general toxicity features of the central atom depict the circular substructures centered on the atom (MGA substructures). In order to better understand what DTE has learned, it is necessary to clarify the approximate radius of the circular substructures highlighted by the general toxicity features. Therefore, we smashed the compounds in the toxicity data sets into circular substructures based on different radii (1−4) and counted the types of the circular fragments and MGA substructures. Since the results of all data sets are consistent, we only present the results of seven data sets in Table 4, and the results of the other data sets are shown in Table S1. According to the numbers of the substructure types in the data sets shown in Table 4, the number of MGA substructure types is between the circular substructures with the radius 2 and 3, suggesting that the radius of the circular substructures captured by the general toxicity features is between 2 and 3. In order to verify this point, we take 4-phenyl-1-(4-phenylbutyl)-piperidine as an example to explore the changes in the general toxicity feature of the node corresponding to N with the change in the structure. The Pearson correlation coefficient was used to assess the similarity of the general toxicity features with the change in the structure. The single bonds were sequentially replaced by double bonds and C atoms by N atoms, and the similarity of the N's general toxicity features between the original structure and changed structure was calculated. Obviously, as shown in Figure 3, when the radius is not higher than 3, the general toxicity features will change obviously, but when the radius is higher than 3, the general toxicity features will not change anymore, suggesting that the radius of the circular fragments representing the general toxicity features is between 2 and 3.

Since the general toxicity features can be captured through DTE by learning a high number of toxicity tasks jointly, they should be able to be transferred to learn single-task models and retain the advantages brought by MGA. To prove this, we first transferred DTE to single-task RGCN and then developed the toxicity prediction model named Transfer-RGCN for each toxicity task individually. We fixed the parameters of DTE and retrained the parameters of TFP-G and toxicity predictor. The detailed performance of Transfer-RGCN can be seen in Table S2, and the change rates of Transfer-RGCN and MGA relative to RGCN can be seen in Figure 4. Obviously, Transfer-RGCN significantly outperforms RGCN overall. In addition, the change rate of Transfer-RGCN relative to RGCN is consistent with that of MGA relative to RGCN, and the Pearson correlation coefficient is 0.9605. This proves that the promotion advantages of MGA are mainly concentrated in DTE, implying that DTE has indeed captured the general toxicity features by learning multiple toxicity tasks jointly. In order to further determine the capability of DTE to learn general toxicity features, we transferred DTE to two external toxicity tasks (hepatotoxicity and urinary tract toxicity) that were not involved in the training of MGA. As shown in Figure 5, Transfer-RGCN outperforms RGCN on both tasks, further demonstrating that general toxicity features were extracted from DTE and can be transferred to new toxicity tasks.

2.2.2. Does TFP-G Really Generate the Customized Toxicity Fingerprints from the General Toxicity Features toward a Specific Toxicity Task? According to our original design, TFP-G should generate the customized toxicity fingerprints toward a specific task by assigning attention weights. We have demonstrated that DTE can indeed extract general toxicity features, so if the parameters of DTE are fixed, TFP-G should always learn similar toxicity fingerprints, regardless of the structures of the subsequent toxicity predictors. Since DTE is fixed, the toxicity fingerprints are only determined by the parameters of TFP-G. Therefore, we changed the number of the fully connected (FC) layers of the subsequent toxicity classifiers and explored the similarity of the parameters in TFP-G for 31 different toxicity tasks. The similarity between the TFP-G parameters of MGA with different FC layers and those of the original MGA with three FC layers can be seen in Figure 6A. Although the structure of

Table 4. Number of the Substructure Types Based on Different Segmentation Methods

	CPBD	CYP1A2	NR-AR	LC50DM	BCF	LC50	IGC50
circular substructure-1	825	1589	1915	344	443	476	364
circular substructure-2	4918	22,334	18,461	1366	2133	2297	2360
MGA substructure	7837	41,830	33,714	1910	3084	3418	4279
circular substructure-3	9222	61,334	45,527	2299	4293	4485	6180
circular substructure-4	11,856	102,147	68,495	2876	6058	5858	9418

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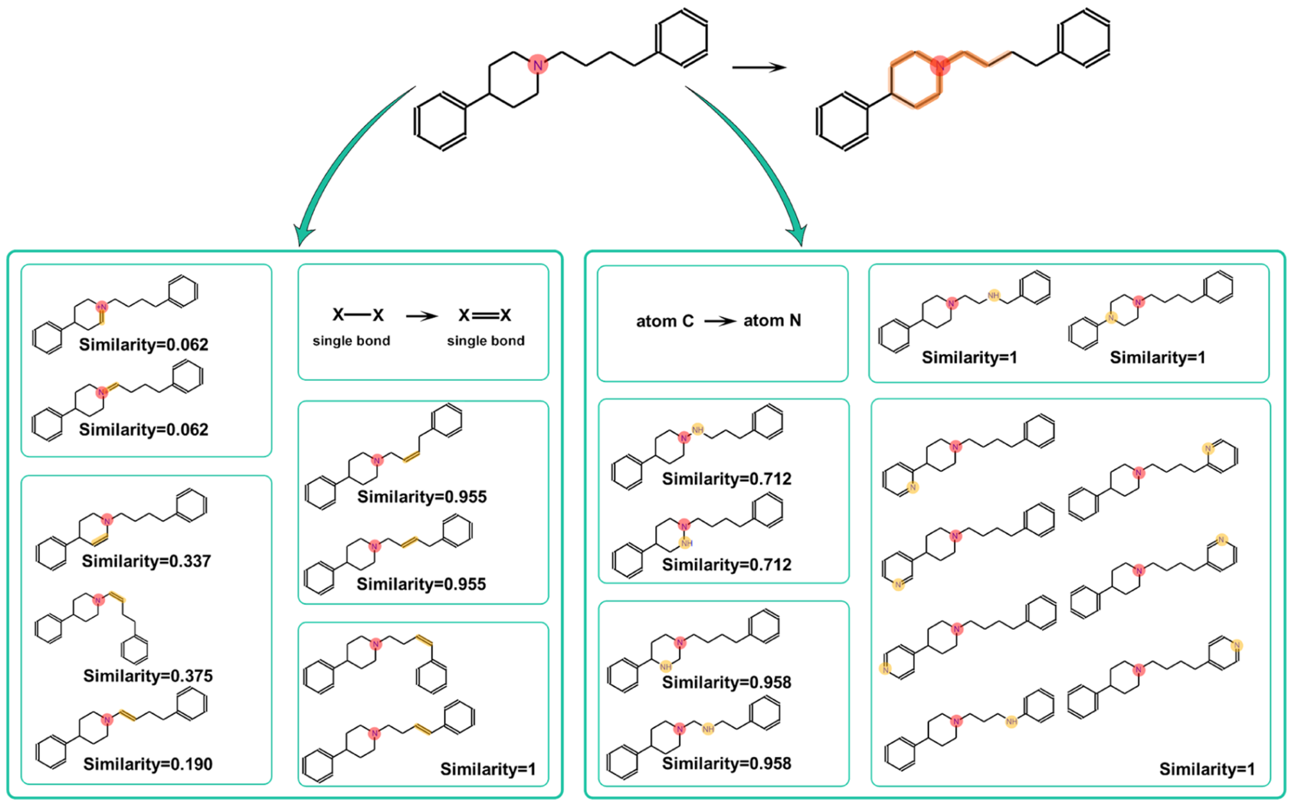

Figure 3. Similarity of the N's general toxicity features with the changes in the surrounding substructures.

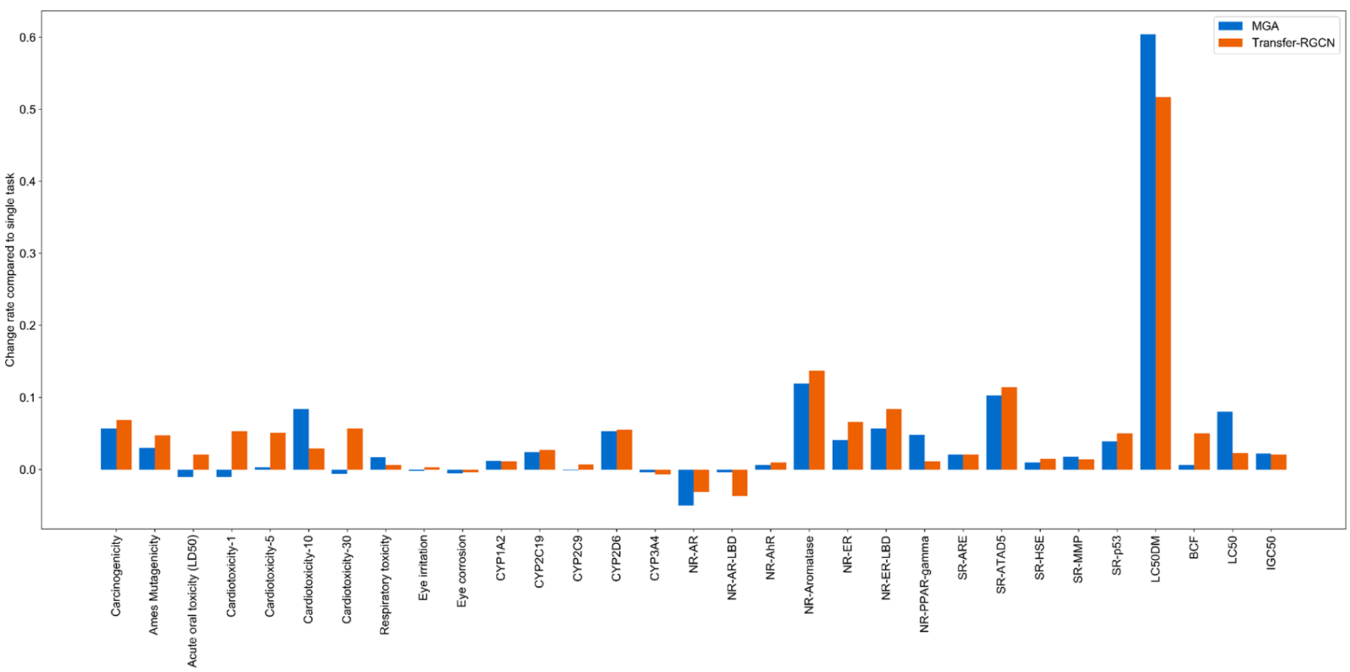

Figure 4. Change rates of Transfer-RGCN and MGA relative to RGCN.

the toxicity predictor is changed, TFP-G can still learn similar toxicity fingerprints, demonstrating that TFP-G can indeed generate the customized toxicity fingerprints for specific toxicity tasks.

Since TFP-G can indeed generate the customized toxicity fingerprints for specific toxicity tasks, it should be able to reveal the relationships between different tasks. Therefore, we analyzed the similarities of the TFP-G parameters between different toxicity tasks. As shown in Figure 6B, the toxicity tasks from the same category tend to have more similar TFP-G parameter similarity compared with other tasks. For example, carcinogenicity and Ames mutagenicity have the most similar TPF-G parameter similarity, EI and EC have the most similar TPF-G parameter similarity, and CYP2C9 and CYP2C19 have the most similar TFP-G parameter similarity. Cardiotoxicity-1, cardiotoxicity-5, cardiotoxicity-10, and cardiotoxicity-30 are

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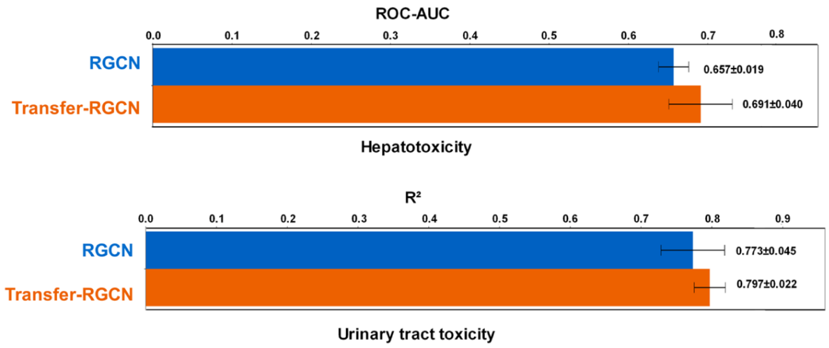

Figure 5. Performances of RGCN and Transfer-RGCN on the hepatotoxicity and urinary tract toxicity tasks.

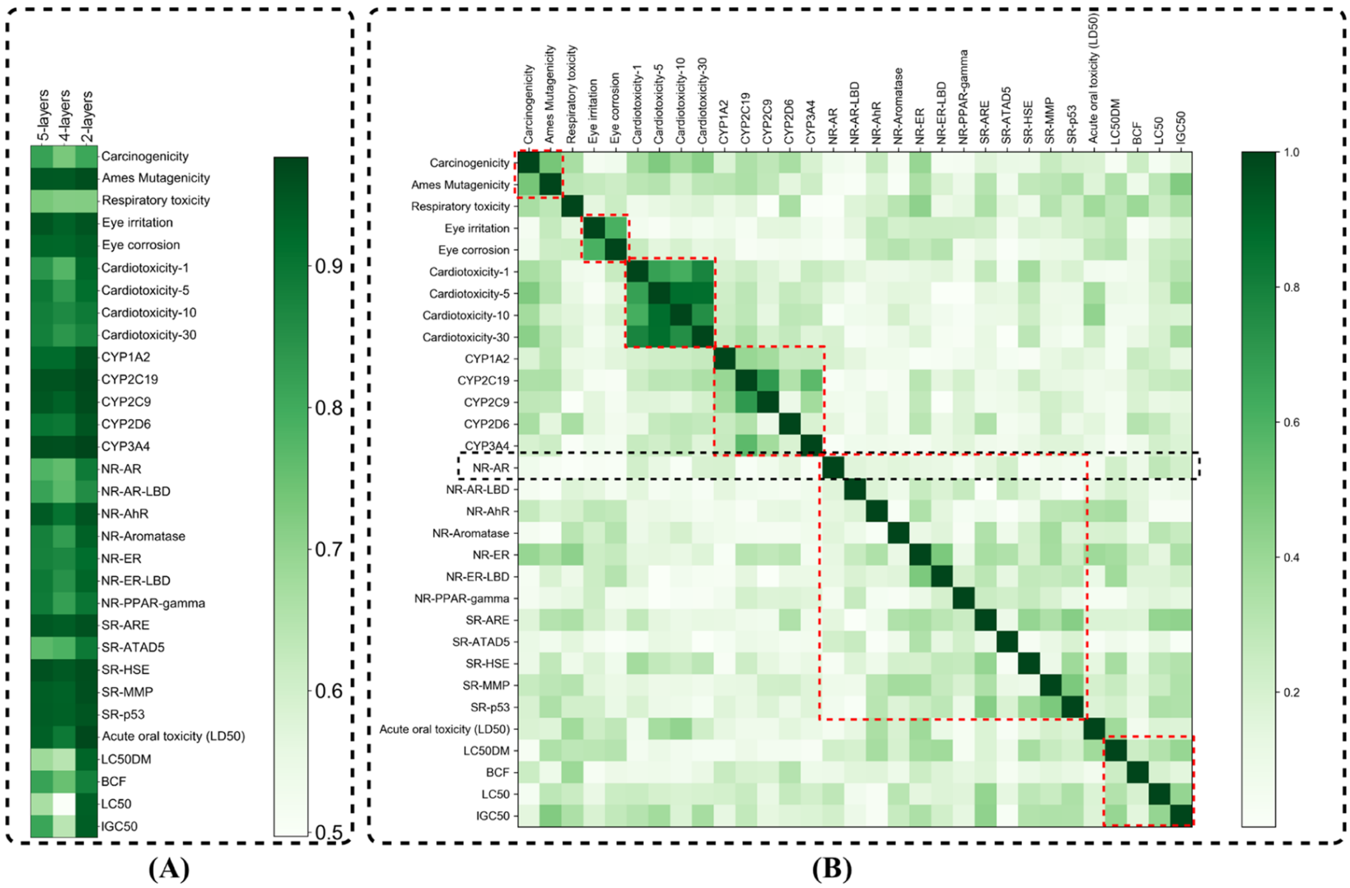

Figure 6. (A) TFP-G parameter similarity between MGA with different FC layers and the original MGA with three FC layers; (B) TFP-G parameter similarities between different toxicity tasks.

the same toxicity endpoint with different classification thresholds. As shown in Figure 6B, the TFP-G parameter similarities between them are obviously stronger than those with other toxicity tasks. In addition, the TFP-G parameter similarities between NR-AR and the other toxicity tasks are quite low, which may be the reason why NR-AR is the only task that has significantly reduced prediction accuracy in MTL. The abovementioned examples further illustrate that TFP-G can indeed generate toxicity fingerprints and reflect the similarities between different toxicity tasks.

Moreover, we compared the modeling capability of the customized toxicity fingerprints with three commonly used molecular fingerprints in QSAR modeling, including ECFP4, RDK, and MACCS. The venerable ML method XGBoost was used to develop the toxicity predictors based on the customized toxicity fingerprints, ECFP4, RDK, and MACCS fingerprints. According to the performances of the different XGBoost models on the test sets shown in Figure 7, the XGBoost models based on the toxicity fingerprints apparently outperform those based on the ECFP4, RDK, and MACCS fingerprints on most tasks, especially on the regression tasks. In some tasks (cardiotoxicity, NR-AR, and NR-AR-LBD) where MGA itself did not perform well, the XGBoost models based on the toxicity fingerprints did not perform so well as well, but they still achieved comparative predictions to those based on other molecular fingerprints. In addition, the dimension of the toxicity fingerprints is 64, which is much smaller than those of the MACCS (167), ECFP4 (1024), and RDK fingerprints (2048), which will greatly reduce the computational consumption in the construction of toxicity prediction models.

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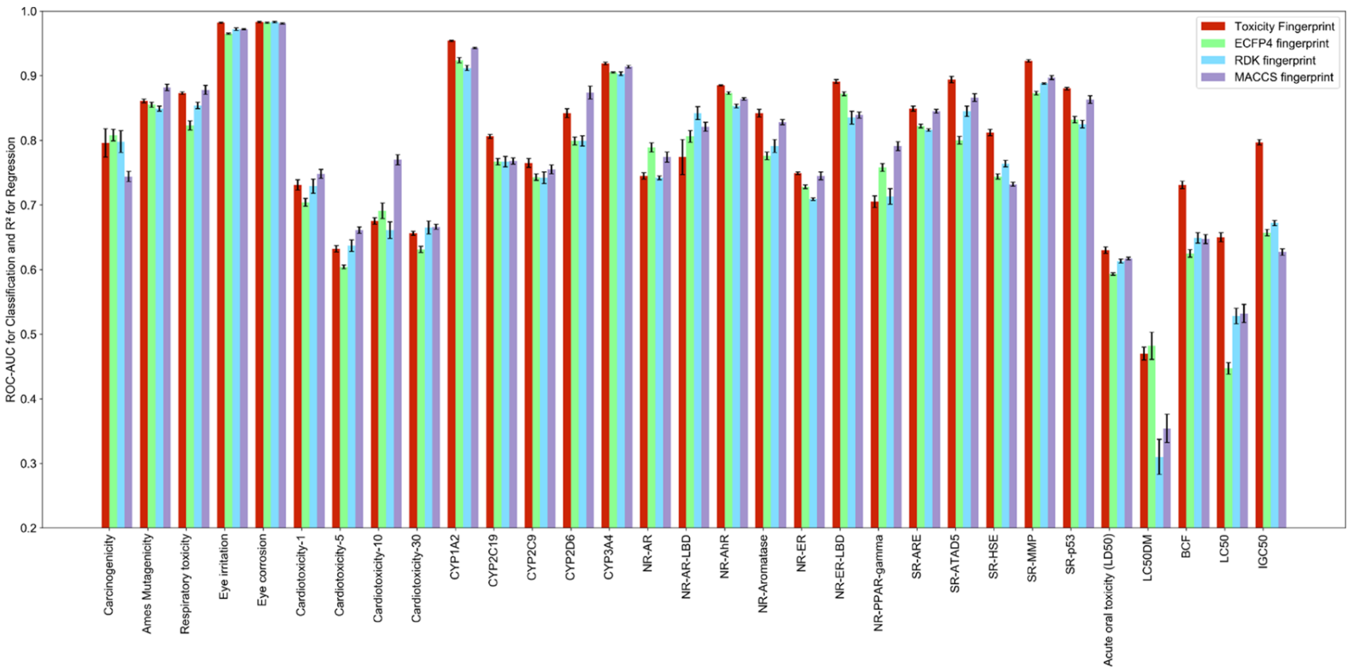

Figure 7. Performances of the XGBoost models on the test sets based on the toxicity fingerprints, ECFP4, RDK, and MACCS fingerprints.

All in all, TFP-G in our method is indeed consistent with our original design, which can generate the customized toxicity fingerprints for specific toxicity tasks. In addition, the parameters of TFP-G can also reveal the relationships between different toxicity tasks, which may help medicinal chemists to better mine toxicity information from related toxicity tasks. Moreover, the toxicity fingerprints generated by TFP-G can be used to build high-precision toxicity prediction models using other ML algorithms, not limited to neural networks.

2.3. Relationships between Substructures and Chemical Toxicity. Understanding the relationships between substructures and chemical toxicity is a concerned issue for medicinal chemists, and our method provides a new way to reveal these relationships. As mentioned above, DTE can extract the general toxicity features from molecules for specific substructures, and TFP-G can generate the customized toxicity fingerprints for specific tasks based on the general toxicity features. When TFP-G generates toxicity fingerprints, different attention weights are assigned to different substructures. We visualized some molecules and their highest-weighted substructures in the Ames mutagenicity and acute oral toxicity test sets and found that the highest-weighted substructures of MGA are consistent with the toxicity SAs reported in the literature.

For example, as reported by Xu et al., acylchloride, semicarbazone, nitrite, nitrosamide, and azide are the SAs according to their high appearance in mutagens and nonmutagens.[35] We visualized some molecules in the Ames mutagenicity test set that contain the abovementioned structures. As shown in Figure 8, the results show that MGA does give the highest weights to these SAs. In addition, alkylfluoride, phosphonic acid derivative, phosphoric trimester, and phosphoric acid derivative are the SAs of acute oral toxicity.[23] We also visualized some molecules in the acute oral toxicity test set that contain the abovementioned structures. As shown in Figure 9, MGA does give the highest weights to these SAs. Whether it is a regression or classification task, MGA can detect the SAs in molecules. Therefore, MGA can identify the customized toxicity SAs for different tasks, which will provide great help for medicinal chemists to explore the relationships between substructures and compound toxicity.

3. CONCLUSIONS

In this study, we proposed the MGA framework to make full use of different toxicity data sets, which can simultaneously develop regression and classification tasks for toxicity predictions. MGA was used to develop a model to comprehensively predict chemical toxicity and reveal the relationships between substructure and chemical toxicities. The predictive power of MGA was highlighted by the excellent predictions to the 31 data sets. We explored if MGA can extract the general toxicity features of circular substructures, and we found that the radius of the circular substructures captured by general toxicity features is between 2 and 3. In addition, we transferred DTE to two external tasks, and the results show that it can improve the prediction accuracy, suggesting that DTE has indeed learned general toxicity features and can be transferred to new toxicity tasks. Furthermore, the customized toxicity fingerprints generated from general toxicity features can be used to build high-precision toxicity prediction models based on ML algorithms, not limited to neural networks. Moreover, the TFP-G parameter similarity can be used to reveal the relationship between different toxicity tasks, which may help medicinal chemists to better mine toxicity information from related toxicity tasks. The relationship between substructures and chemical toxicity is one of the most concerned issues for medicinal chemists. By assigning different attention weights to different substructures, MGA can detect toxicity SAs toward different tasks, which will provide great help for medicinal chemists to mine toxicity information from toxicity data sets.

4. EXPERIMENTAL SECTION

4.1. Collection of Data Sets. In order to build a comprehensive toxicity predictor, we collected 33 data sets covering 10 toxicity categories following Yang's review on in silico toxicity prediction.[5]

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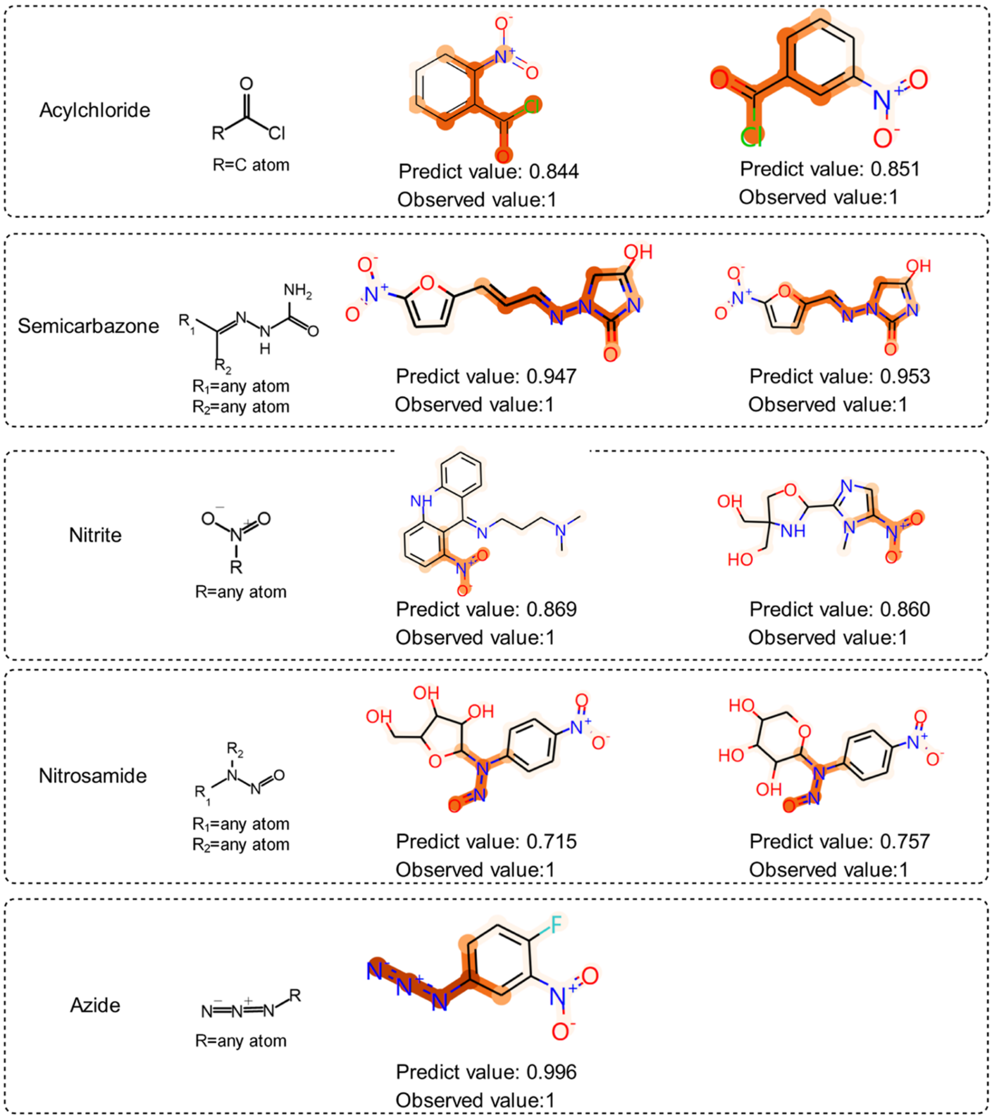

Figure 8. MGA SAs of Ames mutagenicity.

The toxicity categories include carcinogenicity and mutagenicity, acute oral toxicity, cardiotoxicity, respiratory toxicity, irritation and corrosion, CYP450 (Cytochrome P450), endocrine disruption, ecotoxicity, hepatotoxicity, and urinary tract toxicity.[5] The detailed information of the toxicity data sets is summarized in Table 2. The original data sets collected from the literature were revised. The duplicated molecules and the molecules with conflicting label values were excluded. The data sets were split according to the original literature, and the data sets without the clear description of split in the original literature were split by 8:1:1. The specific split results are shown in Table S3, and the detailed description of split is shown in Table S4.

4.2. Methods. *4.2.1. Definitions of Initial Atom and Bond Features.* In this study, molecules were transformed to molecular graphs using the Deep Graph Library (DGL) package, and the molecular graphs were used as the inputs of MGA. A molecular graph is encoded by a set of atom features and bond features. As shown in Table 1, nine types of atom features and five types of bond features are used as the input representation of MGA to characterize atoms, bonds, and their local environment. All atomic information is in one-hot form except that the formal charge and radical electrons are in the integer form. Since the bond information is represented by different categories in MGA, each bond feature will be converted into a specific category (integer) based on five types of bond features.

4.2.2. MGA Framework. Since MTL can make full use of the information learned from related toxicity data sets, we proposed the MGA framework to simultaneously learn the regression and classification tasks for toxicity predictions. Traditional multitask GNN methods usually handle homogeneous tasks, such as pure regression or classification tasks. However, in toxicity prediction, both regression tasks and toxicity tasks are needed. In addition, the traditional multitask GNN always generates multiple predicted values in the output layer, which is extremely difficult to be understood. It is not clear which parts of the model correspond to which tasks. In toxicity prediction, a good method needs to have good interpretability, and thus, it can be used to explore the relationships between substructures and chemical toxicities. To solve the abovementioned limitations of the current multitasking GNN, we proposed an easy-to-interpret framework named MGA to simultaneously learn regression and classification tasks for toxicity predictions.

In terms of functions, MGA can be divided into four components: input, DTE, TFP-G, and toxicity predictor. The detailed architecture of MGA is shown in Figure 1A.

4.2.2.1. Input. The input of MGA is molecular graphs represented by atom features and bond features.

4.2.2.2. Deep Toxicity Extractor. DTE is composed of two GNN layers and can extract general toxicity features from substructures. To ensure that DTE extracts the general toxicity features of circular substructures, we make DTE become the only shared part of MGA. In

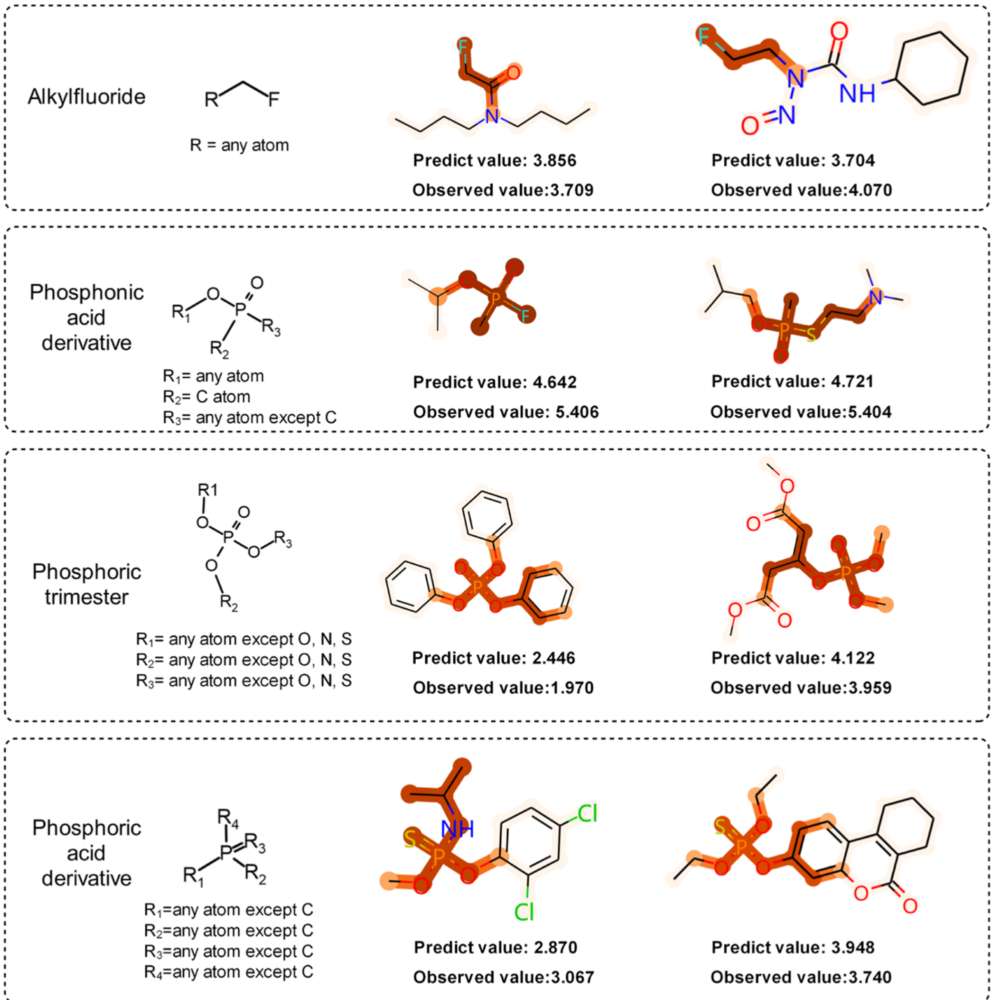

Figure 9. MGA SAs of acute oral toxicity.

the input, a node represents the information of an atom, and after passing DTE, the node represents the circular substructure centered on the atom. In recent years, the GNN has attracted increasing attention in molecular property predictions due to its strong predictive power on graph-structured data and its interpretability.[24−28] All GNN variants, such as GCN,[25] graph attention network (GAT),[58] attentive FP,[24] and D-MPNN[59] can be regarded as the special cases of a simple differentiable message-passing framework. RGCN is an extension of the standard GCN by introducing edge features to enrich the messages used to update the hidden states in the network and has achieved an excellent performance in link predictions and entity classification.[60] Therefore, we used two RGCN layers to build our DTE, and it is worth mentioning that our DTE can also contain other graph network layers. The propagation rule for each node ν is calculated via

$$h_{\nu}^{(l+1)} = \sigma\left(\sum_{r\epsilon R}\sum_{u\epsilon N_{\nu}^{r}} W_{r}^{(l)}h_{u}^{(l)} + W_{0}^{(l)}h_{\nu}^{(l)}\right) \quad (1)$$

where $h_{\nu}^{(l+1)}$ is the state vector of target node ν after $l + 1$ iterations and N_{ν}^{r} denotes the neighbors of node ν under the relation (edge) $r \epsilon R$. In addition, the neighbors of node ν refer to the nodes directly connected to node ν. $\sigma(\,)$ is an element-wise activation function, and ReLU( ) is adopted in DTE. The $W_{r}^{(l)}$ is the weight for neighbor node u connecting to node v by an edge attributed with the relation $r \epsilon R$, and $W_{0}^{(l)}$ is the weight for target node ν. As can be seen above, the edge information is explicitly incorporated in an RGCN under the relation $r \epsilon R$. The weight $W_{r}^{(l)}$ is a linear combination of basis transformation.

4.2.2.3. Toxicity −Fingerprint Generator. As shown in Figure 1B, TFP-G can assign different attention weights to different substructures and then generate the customized toxicity fingerprints from the general toxicity features for a specific toxicity task. The attention weights and toxicity fingerprints are generated as follows:

$$\omega_{\mathrm{v}} = \mathrm{sigmoid}(W\cdot h_{\mathrm{v}} + \mathrm{bias}) \quad (2)$$

$$\mathrm{TFP} = \sum_{\nu=1}^{N}(\omega_{\mathrm{v}}\cdot h_{\mathrm{v}}) \quad (3)$$

where W and bias are the trainable matrixes of the linear transformations of TFP-G learned in model training, N is the number of nodes (substructures), sigmoid( ) is an activation function that limits the attention weight ω_ν of each node (substructure) between 0 and 1, ω_ν is the attention weight of node (substructure) ν, and h_ν is the general toxicity feature of node (substructure) ν.

4.2.2.4. Toxicity Predictor. As shown in Figure 1A, the toxicity predictor predicts the corresponding toxicity based on the customized toxicity fingerprints. The regression and classification toxicity tasks adopt different loss functions (loss_c and loss_r) as follows

$$\mathrm{loss\_}c = \sum_{n=1}^{N}\sum_{c=1}^{C}(-[p_c y_{n,c}\cdot\log\sigma(x_{n,c}) + (1 - y_{n,c})\cdot \log(1 - \sigma(x_{n,c}))]) \quad (4)$$

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$$\text{loss\_}r = \sum_{n=1}^{N}\sum_{r=1}^{R}(x_{n,r} - y_{n,r})^2 \quad (5)$$

where $x_{n,c}$ is the predicted value of molecule n for classification task c, $y_{n,c}$ is the true values of molecule n for classification task c, p_c is the weight of positive samples (toxic molecules), and sigmoid( ) is adopted as the σ( ) in the toxicity predictor. $x_{n,r}$ is the predicted value of molecule n for regression task r, $y_{n,r}$ is the true value of molecule n for regression task r, N is the number of molecules, C is the number of the classification tasks, and R is the number of the regression tasks.

The loss function of MGA is a combination of loss_c and loss_r

$$\text{loss} = \text{loss\_}c + \text{loss\_}r \quad (6)$$

4.2.3. Model Construction and Evaluation. The MGA model was developed using the DGL package with the pytorch backend. The XGBoost models were developed using the xgboost package in python. The detailed hyperparameters of the MGA and XGBoost models are summarized in Table S5. The toxicity data sets were split according to the original literature, and the data sets with no clear description of the split in the original literature were random split by 8:1:1. The predictions on the external test sets were used to evaluate the predictive power of MGA. Moreover, the 31 data sets covering 8 toxicity categories were used to develop the MGA model, while the hepatotoxicity and urinary tract toxicity data sets were used to assess whether the general toxicity features learned by MGA are transferable. The regression tasks were evaluated by the square determination coefficient (R^2), and the classification tasks were evaluated by the area under the receiver operating characteristic curve (ROC-AUC). The source code and implementation of MGA and the data sets can be found at https://github.com/wzxxxx/MGA.

■ ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge at https://pubs.acs.org/doi/10.1021/acs.jmedchem.1c00421.

Number of the substructure types based on different segmentation methods; performances of Transfer-RGCN on the external test sets; split results of the 33 toxicity data sets; detailed description of split; and hyperparameters of each model (PDF)

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Notes

The authors declare no competing financial interest.

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■ ABBREVIATIONS

MGA, multitask graph attention; ML, machine learning; MTL, multitask learning; TL, transfer learning; SA, structural alerts; GNN, graph neural network; CYP450, Cytochrome P450; DGL, deep graph library; DTE, deep toxicity extractor; TFP-G, toxicity fingerprint generator; GCN, graph convolutional network; GAT, graph attention network; RGCN, relation graph convolution network; R^2, square determination coefficient; ROC-AUC, area under the receiver operating characteristic curve; EI, eye irritation; EC, eye corrosion; EDCs, endocrine disrupting chemicals; NRs, nuclear receptors; ER, estrogen receptor; AR, androgen receptor; TEST, Toxicity Estimation Software Tool software; LC50, 96 h fathead minnow LC_{50}; LC50DM, 48 h daphnia magna LC_{50}; IGC50, 40 h tetrahymena pyriformis IGC_{50}; BCF, bioconcentration factor; ECFP, extended connectivity fingerprints; FC, fully connected

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